

## Agenda

- This Week
  - Class 8 – September 23<sup>rd</sup> (2:00– 3:00, 3:30 – 4:30)
  - Class 9 – September 24<sup>th</sup> (10:30 – 12:30)
  - Class 10 – September 25<sup>th</sup> (1:00 – 3:00)
- Starter Quiz
- Core Banking Platform Engineering : Key Generic Functional Objects
  - Product
  - Pricing
  - Account
  - Transaction
  - Balance
  - Accounting
  - Customer Reporting
  - Bank Reporting
  - Regulatory Reporting
- Thought Paper Presentation - Stablecoin (Team 4)

### Note:

- Mobile Phones – Do not bring to class or Completely Switch off
- If students have any specific topic they want to discuss, share the topic and we can try to schedule
- Please be seated by your teams

## Starter Quiz

- Name 3 Indian FinTech Unicorns

Unicorn

Soonicorn

## Payment



## Lending Tech



## Fintech Infra / SaaS



## Starter Quiz

- Let us say, you have taken a loan of 15Lakhs for a car for 3 years at an interest rate of 9%. After 6 months, you get a bonus of 5Lakhs, and you want to pre pay the loan partially.
  - What are the 2 Options that you have and what is the impact of each of the Option

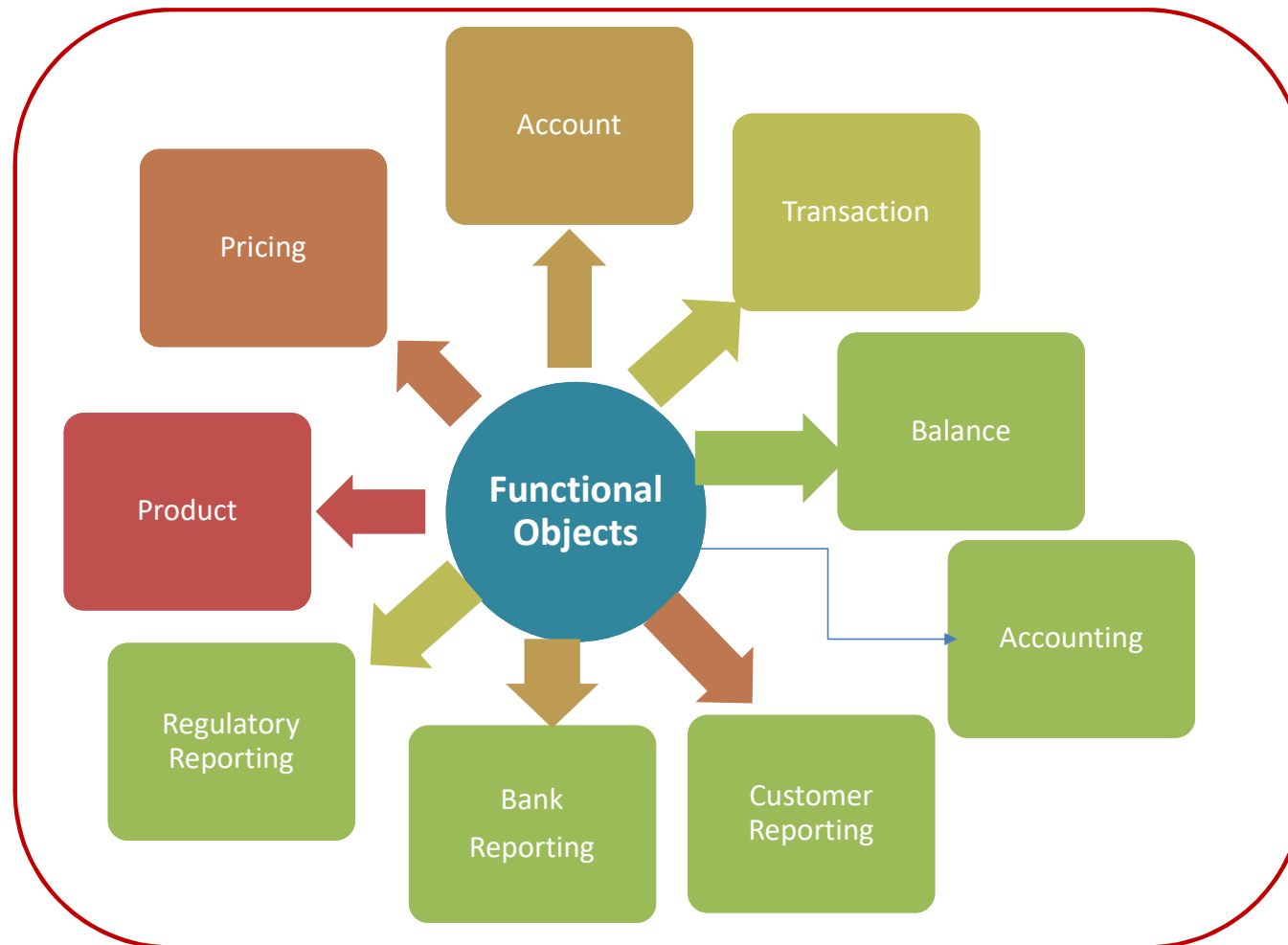
## EMI Vs loan tenure reduction – Factors and Impact Considerations

| Factor                     | Reducing EMI                                                        | Reducing Loan Tenure                                            |
|----------------------------|---------------------------------------------------------------------|-----------------------------------------------------------------|
| <b>Loan Closure</b>        | Takes longer, as tenure remains the same.                           | Faster loan closure due to reduced repayment period.            |
| <b>Financial Relief</b>    | Provides immediate monthly relief by lowering EMI.                  | Offers long-term relief by reducing total interest paid.        |
| <b>Interest Savings</b>    | Moderate savings, as tenure is unchanged.                           | Higher savings on interest because loan ends earlier.           |
| <b>Monthly Commitments</b> | Lower monthly outgo, easier on cash flow.                           | Monthly outgo remains same, but for a shorter duration.         |
| <b>Ideal For</b>           | Borrowers seeking short-term budget flexibility.                    | Borrowers aiming to become debt-free sooner and save more.      |
| <b>Cash Flow Impact</b>    | Improves monthly liquidity, helpful in uncertain income situations. | No monthly relief, but significant long-term financial benefit. |

# Core Banking Platform Engineering – Key Generic Functional Objects

## Key Generic Functional Objects in any Banking Product

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## Product

### Banking Product or Service that is Offered to the Customer

- **Key Attributes:**
  - **Basic Details:** Name, Code
  - **Currency:** List of Currencies the Product is Offered with
  - **Business Rules:** Rules to be adhered to by each Process (e.g. Account Opening/Eligibility, Account Closure etc..)
  - **Transactions:** List of Transactions Allowed and from what Channels/Apps
  - **Charges:** Charges that can be applied for accounts on this Product, this can be Interest , Fees, Commission, Penalties, Bonus, Cash Back ..
  - **Role Types:** List of Role Types available on this product, who and how many can play that role (e.g. Owner, Co Owner, Borrower, Guarantor, Signatory, Beneficiary ...)
  - **Customer Reporting:** List of Reporting / Communication Types available (Alerts, Notices, Statements) and which Role Types it will be sent to
  - **Operational Reporting:** List of Operational Reports Applicable for this Product
  - **Regulatory Compliance & Reporting:** List of Regulations Applicable for this Product
  - **Accounting:** Asset or Liability ( Use case of change from Liability to Asset or vice versa)
- **Change Effectivity:**
  - Current Date – Rarely (only for urgent changes)
  - Future Date – Mostly
  - Backdate – No
- **Change Applicability:** New Accounts, New and Existing Accounts
- **Maker/Checker:** Always
- **Key Related Enterprise Components:**
  - **Enterprise Product Catalog** – Full Catalog of all Products across all Product Lines





## Pricing

### Charge Applied for the Banking Product and Service Provided

- **Key Attributes:**
  - ***Patterns:***
    - Accrual (e.g. Interest)
      - Index Rate Based – Single, Multiple
      - Non Index Rate Based
    - Periodic (e.g. Account Maintenance Fee, Locker Fee)
    - Event Based (e.g. ATM Withdrawal Fee, RTGS/NEFT Branch Initiation fee)
      - Immediate, Deferred
    - One Time (e.g. Loan Processing Fee)
  - ***Charge Applying:***
    - Paid to Account (Deposit Interest, Cash Back, Bonus)
    - Paid from Account (Debit/Lending Interest, Fees, Commission, Penalty)
  - ***Accounting:***
    - Income , Expense
- **Change Effectivity:**
  - Current Date – Rarely (only for urgent changes)
  - Future Date – Mostly
  - Backdate – Rarely (only Interest Rate Changes, Processing Impact)
- **Change Applicability:** New Accounts Only , New and Existing Accounts
- **Maker/Checker:** Always
- **Key Related Enterprise Components:**
  - **Enterprise Pricing** – Pricing determination made not at a single Product/Account Level but at the Overall Customer Relationship level covering all Products or Bundling/Package Products

# Account

## Financial Account created for a customer when a Banking Product is Sold

- **Key Attributes:**
  - **Basic Details:** Account Name, Title
  - **Currency:** Currency of the account (Single)
  - **Business Rules:** Customized Business Rules if allowed on the specific account
  - **Charges:** Customized Charges if allowed on the account (Deviation from Product or Fixed Value)
  - **Customers/Involved Parties Related:** Customer / Involved Parties related and what is the Role they Play on the account (e.g. Owner, Co Owner, Borrower, Guarantor, Signatory, Beneficiary ...)
  - **Customer Communication Types:** List of Communication Types available (Alerts, Notices, Statements) and which Role Types it will be sent to
- **Change Effectivity:**
  - Current Date – Mostly
  - Future Date – Rarely (For Selected Type of Changes)
  - Backdate – Rarely (and only Interest Rate Changes)
- **Maker/Checker:** Conditional (Based on product or customer classification and type of information being changed)
- **Key Related Enterprise Components:**
  - RDS (Reporting Data Store) and EDW (Enterprise Datawarehouse)



Transaction

## Financial Transaction on an Account

- **Key Attributes:**
  - **Initiation:**
    - Customer Initiated (Deposits, Withdrawals, Transfers, Payments ...)
    - Platform Initiated (Interest, Fees ..)
    - 3<sup>rd</sup> Party Initiated (Salary Payments, Govt Subsidies, Dividends, Tax Refunds ..)
  - **Effectivity:**
    - Current – Mostly
    - Backdated – Rarely (Processing Impact)
    - Future Dated - Limited
  - **Frequency:**
    - Single
    - Recurring
  - **Occurrences**
    - Single
    - Bulk (e.g. Clearing Payments, Salary Payments, Govt Payments.)
  - **Currency:**
    - No Currency Exchange
    - Currency Exchange Needed
  - **Change Handling:**
    - Reversal
    - Adjustment
  - **Maker/Checker:** Conditional based on Amount or Account Classification / Status

## Financial Transaction on the Account

- **Key Attributes:**
  - **Business Rules:** Vary by Product and Channel
  - **Balances:** Affects 1 to many Balance Types
  - **Charges:** Event based charge Immediate or Deferred
  - **Accounting:** Generates GL Accounting Entries that need to Balance
  - **Customer Reporting:** Alerts or Notices or Statements
  - **Transaction References:**
    - Internal – System Generated
    - External (1:Many) – Used for Transaction Reconciliation between Systems
  - **Exception Processing:** Need a Robust Exception Processing capability along with the related reporting to handle Transaction exception logging and actioning
- **Global Industry Standards:**
  - **ISO 8583:** ATM/POS/Cards
  - **ISO 20022:** Financial Messaging Standard
- **National Standards:**
  - Vary by Country: e.g. UPI/IMPS/NEFT/RTGS in India, ACH/Fed Now/Fedwire's in US
- **Key Related Enterprise Components and Payment Apps:**
  - Enterprise Payments Hub
  - ATM/POS
  - Payment Networks and Payment Aggregators
  - RDS (Reporting Data Store)

Transaction

# Balance

## Balances (Account Balances)

- **Key Attributes:**
  - ***Change Model:***
    - Event / Transaction Driven
      - Financial Transaction, Overdraft Limit Setup/Updates ..
    - Period Driven
      - Period Duration: Daily, Monthly, Quarterly, Yearly (Financial Year), Life to Date ..
      - Values in the Period: Minimum, Maximum, Average
  - ***Computation Model:***
    - Simple Addition/Subtraction e.g.
      - $\text{Ledger Balance} = \text{Sum of all Debits and Credit transactions done on the account}$
    - Formula Driven e.g.
      - $\text{Available Balance} = \text{Ledger Balance} + \text{Overdraft Limit} - \text{Hold Balance} - \text{Float Balance}$
  - ***Persistence Model:***
    - Based on the need Balances can be Computed & Persisted or only Computed only
    - E.g. Ledger Balance is Computed & Persisted, Available Balance is Computed only
  - ***Usage:***
    - Transaction Validation
    - Charging (Interest or fee Basis Balance, Threshold Balance ..)
    - Reporting (Customer, Operational, Regulatory) & Analytics
- **Key Related Enterprise Components:**
  - RDS (Reporting Data Store) and EDW (Enterprise Datawarehouse)

### Note:

- Account Counters tracking of number of occurrences of an event / transaction follows a similar principles as above
- Customer Balances, Branch Balances are not covered here

## Accounting

- **Key Attributes:**
  - ***Accounting Period:***
    - Financial Year Period such as January 1<sup>st</sup> to December 31<sup>st</sup> or April 1<sup>st</sup> to March 31<sup>st</sup>
    - Assets, Liability, Income, Expense are aggregated for this Period
  - ***Accounting Model:***
    - Provides the rules of how to account and report
    - Most Commonly used base models are GAAP, IFRS and each country has its own model refining a base model
  - ***Accounting Classifications:***
    - Assets, Liabilities, Income, Expense
    - 5<sup>th</sup> is Equity not much applicable from a Core Banking Platform perspective
  - ***GL (General Ledger) Account:***
    - GL accounts are used to generate financial statements and provide a summary of a Bank's financial position
    - All the GL accounts used by a Bank are listed under the Chart of Accounts
    - GL accounts are the detailed line items, and accounting classifications are the broad groupings that organize these items for financial reporting purposes.
- **Key Related Enterprise Components:**
  - **Enterprise General Ledger:** Aggregates Information from all Core Product Processors and does financial reporting at the overall Bank Level

## Customer Reporting

- **Key Attributes:**
  - ***Reporting Triggers (Automated): When to Report***
    - Current Event / Transaction: When an Event / Transaction occurs
    - Upcoming Event / Transaction: FD Maturity / Renewal, Loan Repayment
    - Period Completion: Statement Period, Financial Year
    - Past Due: Past Loan Repayment Date
  - ***Reporting Patterns: How to Report***
    - Alerts:
      - Near Real Time
    - Notices (or Bills in Lending)
      - Near Real Time
      - Day End
    - Statements – Single or Combined
      - Day End
      - Near Real Time (On Demand by Customer or Bank Staff)
  - ***Recipient: Whom to Report***
    - Role Types: Owner, Co Owner, Borrower, Guarantor, Signatory etc..
  - **Key Related Enterprise Components:**
    - **Output Rendering Component:**
      - Formatting (Stylesheet, Language, Logo ..)
      - Distribution (SMS, WhatsApp, E Mail, Print and Post)

## Bank Reporting

- **Key Attributes:**
  - ***Reporting Trigger: When to Report***
    - Product/Pricing Changes or High Value/Critical Account Changes done/become effective
    - Day End
    - Adhoc (On Demand)
  - ***Reporting/Communication Patterns: How to Report***
    - Alerts: Banker Alerts
      - Near Real Time
    - Reports
      - Scheduled Reports (Typically End of Day): Key Categories:
        - Trial Balance and System Balance Reports
        - Account Activity and Audit Reports
        - Exception Reports ...
      - Real Time Reports ( Generated on Demand by User)
- ***Recipient: Whom to Report***
  - Product Managers – For any Product Changes
  - Relationship Managers of High Value / Critical Customers for any Critical Account Activity or High Value Transactions
  - Branch / Audit Staff if any activity is attempted to be done on Restricted Accounts
  - Bank Operations Staff
- **Key Related Enterprise Components:**
  - **RDS (Reporting Data Store):**
    - Formatting (Stylesheet, Language, Logo ..)
    - Distribution (SMS, WhatsApp, E Mail, Print and Post)



## Regulatory Reporting

- **Key Attributes:**
  - ***Reporting Trigger: When to Report***
    - Frequency, Format, Content as dictated by the Regulator
  - ***Reporting/Communication Patterns: How to Report***
    - Reports
    - Extracts
  - ***Recipient: Whom to Report***
    - Regulator – Country Regulator ('s)
    - Bank - Compliance Team
- **Key Related Enterprise Component:**
  - **Regulatory Reporting Component: Country Specific**
    - Creates Reports (and Extracts) in the Format and Content as specified by the Regulatory Authority